



CNR

CRAFT N ROAM

AI Travel Decision Simulator & 3D Walk

Craft a transparent plan. Roam the winner in 3D.

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Generative AI · Final Project Presentation

The Problem

Trip planning means balancing budget, time, and taste across dozens of options — and every existing tool skips the reasoning.

Listing sites

Google Maps, TripAdvisor, Yelp show everything and decide nothing.

AI chatbots

One suggestion, no visible tradeoffs — a black box.

Either way

No way to get a real feel for a place before you commit.

Craft, Then Roam

A transparent, traveler-adjustable decision engine, followed by a review-grounded 3D preview of the winner.

1. Craft the plan

- Traveler describes their trip in free text (69 tokens)
- LLM parses it into structured weights
- Deterministic engine scores & ranks 5 curated spots
- Sliders let travelers fine-tune live, for free

2. Roam in 3D

- Traveler unlocks all spots in 3D once (89 tokens)
- LLM reads that spot's curated reviews
- Extracts ground / sky / palette / props (fixed catalog)
- Seeded layout builds a walkable Three.js scene

Demo — Craft a Transparent Plan

"A relaxed five-hour trip, mid-range budget, good food"

- Cours Saleya Market scores 0.94 — strong match, perfect cost & time fit
- Sliders adjust budget, time, and category balance
- Re-ranks instantly — no extra tokens, no black box

CNR - Craft N Roam
Craft your trip, get a transparent plan, then roam the top spot in 3D. 231 tokens

[← Start over](#)

"You want a relaxing 5-hour trip with a mid-range budget and good food included."
Cours Saleya Market scored highest (0.94) - strong match on preferences (0.80), cost fit (1.00), and time fit (1.00). Promenade des Anglais was runner-up (0.93) but scored lower on time fit.

Fine-tune your plan
Adjusting these re-ranks instantly - no extra tokens.

Budget level: 3 / 5

Time available: 5 hrs

Relax: 0.80 Adventure: 0.20

Culture: 0.30 Food: 0.80

Cours Saleya Market 94

Demo — Roam the Winner

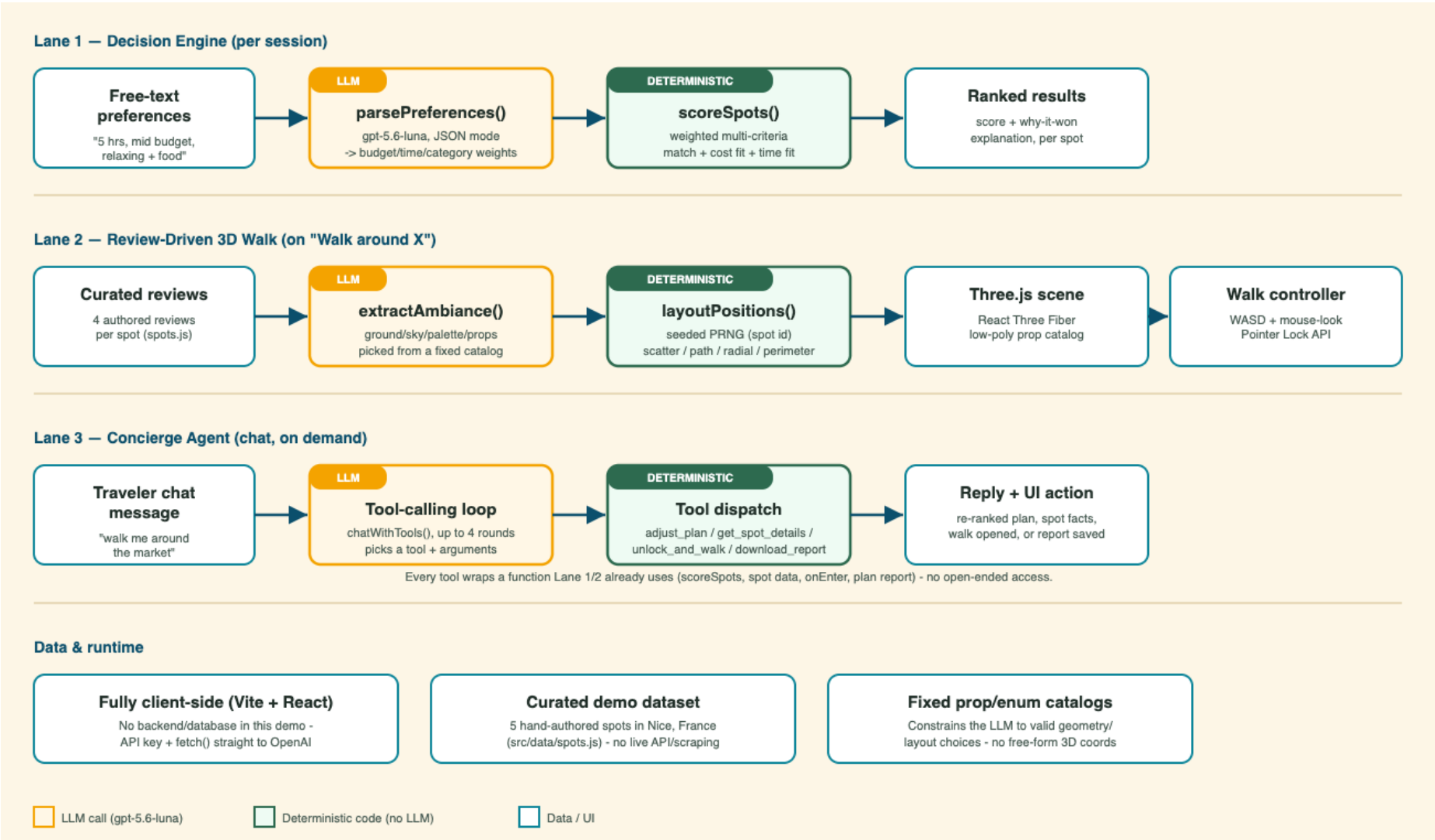
- LLM reads Cours Saleya Market's reviews
- Picks ground, sky, palette, and props from a fixed catalog
- A seeded algorithm places everything — same input, same scene
- Explorable in first person: WASD + mouse-look
- Share 3 photos + a review per spot, earn 30 tokens back



Stylized, not photorealistic — by design (see limitations).

How the AI Is Used

Three lanes of narrow LLM steps; every decision and every agent tool call bottoms out in ordinary deterministic code.



Demo — Ask the Concierge (Lane 3)

A real tool-calling agent - four tools, each a thin wrapper around a function the app already trusted.

- `adjust_plan` - "make it cheaper" re-ranks live
- `get_spot_details` - answers grounded in curated reviews
- `unlock_and_walk` - "walk me around the market" opens the 3D scene
- `download_report` - saves the plan report

Ask the CNR concierge

Try "make it cheaper", "tell me about Castle Hill", or "walk me around the market".

What is the best time to visit Castle Hill and is it crowded?

Castle Hill is best early in the morning—especially around sunrise—for the views and quieter paths. It's generally less crowded than the Promenade, though the summit and waterfall can get busier later; expect a steep walk and uneven steps.

Type a message... Send

Capped at 4 tool-call rounds per turn - no runaway loops.

One Design Principle

Lanes 1-2: the LLM translates, it never decides. Lane 3 may act - but only through tools that wrap trusted code.

LLM calls

- Free text -> structured weights
- Reviews -> structured ambiance spec
- Fixed JSON schema, fixed enum catalogs
- Validated & clamped before use

Deterministic code

- Weighted multi-criteria scoring
- Live plan-editor re-ranking
- Seeded PRNG for 3D layout
- Same inputs -> same outputs, always

A Real Integration Bug We Hit

gpt-5.6-luna rejects function-tool calling combined with its default reasoning_effort on /v1/chat/completions.

```
invalid_request_error - param: reasoning_effort
```

```
"Function tools with reasoning_effort are not supported for gpt-5.6-luna in /v1/chat/completions."
```

Fix: request raw JSON (response_format: json_object) instead of tool-calling, and pass reasoning_effort: "none" explicitly.

One Honest Limitation

The 3D scenes are stylized, low-poly interpretations — not photorealistic reconstructions of the real locations.

True 3D reconstruction needs dozens of calibrated, overlapping photos (photogrammetry / NeRF) — not review text or a handful of casual photos. We chose to be upfront about this rather than overclaim, and built a deterministic, review-grounded stylized scene instead. The token economy is the same story: presented in full, built as local state only — no real payment happens anywhere in this demo.

Also flagged: WASD movement needs a final real-browser verification pass (automated headless testing was inconclusive due to background-tab throttling).

The Business Model (Presentation Only)

Simulated in the demo as local app state — no real payment processing exists today.

Item	Price
Small token bundle	\$0.80 -> 100 tokens
Large token bundle	\$4.00 -> 600 tokens (~17% off)
Generate a plan (search)	69 tokens (~\$0.46-0.55)
Unlock 3D walking (all spots)	89 tokens (~\$0.60-0.71, one-time)
Share 3 photos + a review, per spot	+30 tokens back

Margin check

Actual LLM cost per search: a fraction of a cent. Priced at 69 tokens (~\$0.50), that is roughly two orders of magnitude of margin before hosting, support, and refunds.

What's Next

Headline idea: crowdsourced photos, real rewards

Travelers upload up to 3 of their own photos per spot -> vision model extracts richer ambiance -> traveler earns a real 30-token reward per spot. Fully legitimate (user-submitted, not scraped). Needs accounts, storage, and moderation — a real backend, deliberately out of scope for now.

- Real payment processing and persistent accounts for the token economy
- Move the OpenAI call server-side before any public deployment
- Hotel, flight, and other token-gated booking add-ons
- Expand beyond Nice to more destinations
- Touch controls for mobile (today assumes a keyboard)
- A real usability study outside the two-person team

Thank you

Craft a transparent plan. Roam the winner in 3D.

<https://github.com/GongNT/travel-3d-simulator>

